

120 Days of Quant Finance

Week 6: Stochastic Processes & Dynamic Asset Modeling

Day 36: Random Walks and Discrete-Time Price Models

Arnav Garg

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Introduction

Financial asset prices evolve dynamically through time and cannot be described by single random variables. Instead, they are modeled as stochastic processes, which capture the randomness of price movements across time.

This section introduces discrete-time stochastic processes through random walks and basic price evolution models.

Stochastic Processes

A stochastic process is a collection of random variables indexed by time:

$$\{X_t\}_{t \in T}.$$

Each X_t represents the value of the process at time t . In finance, stochastic processes are commonly used to model asset prices, returns, interest rates, and volatility.

Discrete-Time Framework

In discrete-time models, time advances in fixed intervals:

$$t = 0, 1, 2, \dots$$

Discrete-time modeling aligns closely with observed market data and provides a foundation for more advanced continuous-time models.

The Random Walk Model

A simple random walk is defined as:

$$X_t = X_{t-1} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ are independent and identically distributed random variables with:

$$E[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2.$$

This model captures the idea that future price changes are unpredictable and independent of past movements.

Returns-Based Modeling

Rather than modeling prices directly, financial models typically focus on returns.

Simple returns are defined as:

$$R_t = \frac{S_t - S_{t-1}}{S_{t-1}},$$

while log-returns are given by:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right).$$

Returns are approximately stationary and easier to model statistically.

Discrete-Time Price Evolution

A basic discrete-time price model can be written as:

$$S_t = S_{t-1}(1 + R_t),$$

where returns follow:

$$R_t = \mu + \sigma \varepsilon_t.$$

Here, μ represents the expected return (drift), while σ controls the magnitude of random fluctuations.

Martingale Intuition

A stochastic process $\{X_t\}$ is a martingale if:

$$E[X_t \mid \mathcal{F}_{t-1}] = X_{t-1}.$$

This property implies that the best prediction of tomorrow's value, given all available information today, is today's value.

Conclusion

Random walks provide a fundamental framework for modeling asset price dynamics. Discrete-time stochastic processes form the basis for continuous-time models used in option pricing, risk management, and time series analysis.